Quantitative Research

Part 2: Graphical Representations

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About the Lecturer



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Professor of Mathematics, specialising in Business Data Science

- Former Technical Lead for Artificial Intelligence and supervisor of industrial doctoral projects at BASF
- Research: Data Science, Machine Learning and Artificial Intelligence
- Focus: Application of AI in business and industry

Course materials:
github.com/ChrisW09/Quantitative-Research-Methods

The lecture at a glance

	Part	Content
	1 – Descriptive Statistics	Basic concepts, levels of measurement, frequencies, measures of central tendency and dispersion
▶	**2 – Graphical Representations**	**Empirical CDF, histogram, boxplot**
	3 – Statistical Inference	Normal distribution, CLT, confidence intervals, sample size, hypothesis tests, p-values, one-sided tests, types of error
	4 – Regression	Correlation, least squares, goodness of fit, inference for regression, multiple regression (appendix: residual analysis)
	5 – Classification	Conditional probability, logistic regression, classification rules, confusion matrix, ROC/AUC, Bayes classifiers
	Python course	Separate course: Python basics, pandas, plots, machine learning (Colab notebooks)

All analyses in **Python** (NumPy, pandas, SciPy, statsmodels, matplotlib); every deck comes with a Python script and a Jupyter notebook – also in Google Colab.

Contents – Part 2

1. Empirical Cumulative Distribution Function
2. Histogram
3. Boxplot
4. Exercises
5. Python Commands
6. Summary

▶ **Try it yourself:** notebook, all examples and exercises of this part – open in Colab

Learning objectives – Part 2

After this part you will be able to ...

- construct the **empirical CDF** and read off shares and quantiles from it,
- build a **histogram** with equal and unequal class widths (densities!) and read shares off a given histogram (areas!),
- construct a **boxplot** step by step, including the whiskers and outliers,
- **read and interpret** given graphics (symmetry, skewness, outliers) and compare groups with boxplots,
- combine the numbers of Part 1 with the pictures of Part 2,
- create every graphic in Python (`matplotlib`).

Key takeaway

The running story continues: the **shoe sizes of the soccer team** from Part 1 – now drawn instead of computed.

List of Symbols (1)

Symbol	Meaning	Symbol	Meaning
ω	Statistical unit	h_j	Absolute frequency
n	Number of statistical units	f_j	Relative frequency
X	Variable (characteristic)	H_j	Cumulative absolute frequency
x	Value of the variable	$F(x)$	Empirical CDF
x_i	Observed value	a_j	Class midpoint (class j)
$x_{(i)}$	i-th value of the ordered sample	$\tilde{x}$	Median
$\bar{x}$	Arithmetic mean	$\bar{x}_{\text{geo}}$	Geometric mean

List of Symbols (2)

Symbol	Meaning	Symbol	Meaning
s^2	Sample variance	v	Coefficient of variation
s	Standard deviation	v^*	Normalised coeff. of variation
R	Range	Δx_j	Class width (class j)
IQR	Interquartile range	$f_j/\Delta x_j$	Density (histogram)
$x_{0.25}$, $x_{0.75}$	25 % and 75 % quantile	k	Whisker factor (here 1.5)
Q_1, Q_3	1st and 3rd quartile	x_p	p-quantile $= \min\{x \mid F(x) \geq p\}$
$\lceil \cdot \rceil$	Rounding up	$c_0, \ldots, c_k$	Class boundaries

Overview

1 Empirical Cumulative Distribution Function
- Exercises

2 Histogram

3 Boxplot

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5 Python Commands

6 Summary

Empirical CDF – Construction

Guiding question

What share of the soccer players wear shoe size 39 or smaller? And up to which size do we find at least half of the players?

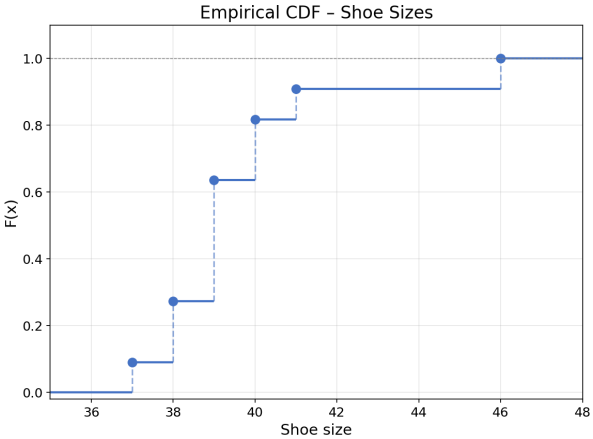
Construction: The observed values are plotted to scale on the x-axis; the corresponding y-values are given by the cumulative relative frequencies. The result is a step function whose y-values lie between 0 and 1.

- The height of each step depends on how often the value was observed
- Higher frequencies $\rightarrow$ taller steps

Frequency Table

Shoe size	37	38	39	40	41	46	Total
Absolute frequency	1	2	4	2	1	1	11
Relative frequency	0.0909	0.1818	0.3636	0.1818	0.0909	0.0909	1
Relative frequency in %	9.1 %	18.2 %	36.4 %	18.2 %	9.1 %	9.1 %	100 %
Relative frequency	$\frac{1}{11}$	$\frac{2}{11}$	$\frac{4}{11}$	$\frac{2}{11}$	$\frac{1}{11}$	$\frac{1}{11}$	1
Cum. rel. frequency	0.0909	0.2727	0.6364	0.8182	0.9091	1	
Cum. rel. frequency	$\frac{1}{11}$	$\frac{3}{11}$	$\frac{7}{11}$	$\frac{9}{11}$	$\frac{10}{11}$	1	

Empirical CDF



Empirical CDF – Definition

Definition (the "empirical distribution function" of Part 1)

$$F(x) = \sum_{x_i \le x} f(x_i) = \frac{\text{number of observations} \le x}{n}$$

(the sum runs over the *distinct* observed values $x_i \le x$)

In words: $F(x)$ is the share of observations that are **at most** x. Example: $F(39) = \frac{7}{11} = 0.6364$ – 63.6 % of the players wear size 39 or smaller.

- Computing the cumulative relative frequency for every observed value yields the empirical CDF
- Between two observed values F stays constant; at each observed value it jumps up by $f(x_i)$
- Quantiles can therefore be read off directly from it

Empirical CDF – Reading Off Quantiles

Example: Median. The median is the smallest value at which the empirical CDF **reaches or exceeds** 0.5:

$$\tilde{x} = x_{0.5} = F^{-1}(0.5) = \min\{x \mid F(x) \geq 0.5\}$$

Shoe sizes: $F(38) = 0.2727 < 0.5$, but $F(39) = 0.6364 \geq 0.5 \Rightarrow \tilde{x} = 39$.

Generalisation

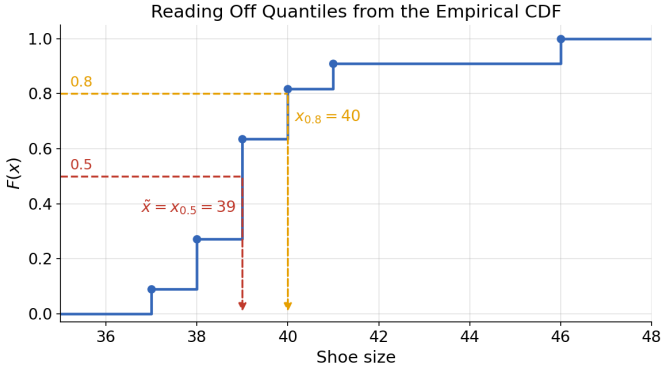
Every p-quantile ($0 < p \leq 1$) can be obtained from the empirical CDF:

$$x_p = F^{-1}(p) = \min\{x \mid F(x) \geq p\}$$

In words: go up to the height p, move right until you hit the step function and read off the x-value there.

Why "$\geq p$" and not "$= p$"? F jumps – here it never takes the value 0.5.

Empirical CDF – Reading Off Quantiles Graphically



80 % quantile: $F(39) = 0.6364 < 0.8 \leq F(40) = 0.8182 \Rightarrow x_{0.8} = 40$.
Same values as with the position rule $x_p = x_{(\lceil n \cdot p \rceil)}$ of Part 1: $\lceil 11 \cdot 0.5 \rceil = 6 \rightarrow 39$ and $\lceil 11 \cdot 0.8 \rceil = 9 \rightarrow 40$.

Empirical CDF in Python

Computing

```
import numpy as np, matplotlib.pyplot as plt
from statsmodels.distributions.empirical_distribution import ECDF
shoe = np.array([40, 39, 38, 41, 46, 39, 39, 38, 37, 40, 39])
F = ECDF(shoe)     # F is a function of x
```

Evaluating

```
F(39)              # value of the empirical CDF at x = 39: 0.6364 (= 7/11)
```

Plotting

```
plt.ecdf(shoe)     # step function incl. vertical jumps (matplotlib >= 3.8)
```

- **Try it yourself:** notebook, section 1 "Empirical Cumulative Distribution Function" – open in Colab

Exercise – Complaints per Day

Problem statement

An online shop records the number of customer complaints per day on $n = 10$ days:

$$2,\ 0,\ 3,\ 1,\ 2,\ 5,\ 1,\ 3,\ 2,\ 6$$

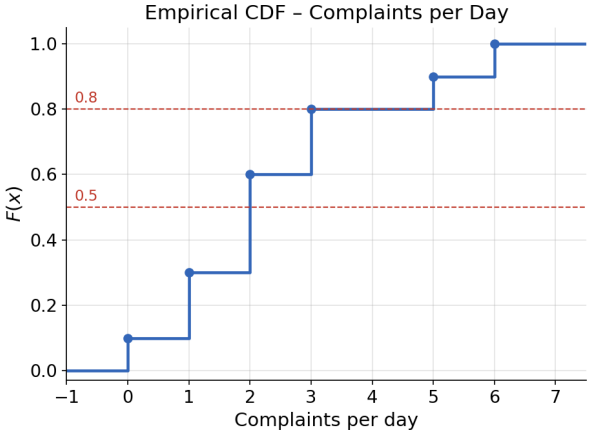
1. Set up the frequency table with absolute frequencies h_j, relative frequencies f_j and cumulative relative frequencies $F(x)$, and draw the empirical CDF.
2. Read off: What share of the days had at most 2 complaints? At most 4 complaints? More than 1 but at most 3 complaints?
3. Determine the median and the 80 % quantile from the empirical CDF.
4. Which Python commands check your results?

Solution (1/4) – Complaints: Frequency Table and Empirical CDF

Sorted: 0, 1, 1, 2, 2, 2, 3, 3, 5, 6

x	h_j	$f_j = h_j/n$	$F(x)$
0	1	0.1	0.1
1	2	0.2	0.3
2	3	0.3	0.6
3	2	0.2	0.8
5	1	0.1	0.9
6	1	0.1	1.0
Total	10	1	

Rule: $F(x) = \sum_{x_i \le x} f(x_i)$ – at every observed value F jumps up by f_j.



Solution (2/4) – Complaints: Reading Off Shares

Rule: $F(x)$ is the share of observations that are **at most** x. The share in the interval $(a, b]$ is the difference $F(b) - F(a)$.

At most 2 complaints

$F(2) = 0.1 + 0.2 + 0.3 = \mathbf{0.6} \rightarrow$ on 60 % of the days there were at most 2 complaints.

At most 4 complaints

The value 4 was never observed – the step function stays constant between 3 and 5:
$F(4) = F(3) = \mathbf{0.8} \rightarrow$ 80 % of the days.

More than 1 but at most 3 complaints

$F(3) - F(1) = 0.8 - 0.3 = \mathbf{0.5} \rightarrow$ on half of the days there were 2 or 3 complaints.

Solution (3/4) – Complaints: Reading Off Quantiles

Rule: $x_p = \min\{x \mid F(x) \geq p\}$ – the first value at which the step function reaches or exceeds the height p.

Median ($p = 0.5$)

$F(1) = 0.3 < 0.5$, but $F(2) = 0.6 \geq 0.5 \rightarrow \tilde{x} = x_{0.5} = \mathbf{2}$ complaints.

80 % quantile ($p = 0.8$)

$F(2) = 0.6 < 0.8$, but $F(3) = 0.8 \geq 0.8 \rightarrow x_{0.8} = \mathbf{3}$: on at least 80 % of the days there were at most 3 complaints.

Check with the position rule $x_{(\lceil n \cdot p \rceil)}$

Median: $\lceil 10 \cdot 0.5 \rceil = 5 \rightarrow$ 5th sorted value $= 2$. 80 %: $10 \cdot 0.8 = 8$ is already an integer $\rightarrow$ position $8 \rightarrow 3$.

Solution (4/4) – Complaints: Python

Shares, quantiles and plot

```
from statsmodels.distributions.empirical_distribution import ECDF
compl = np.array([2, 0, 3, 1, 2, 5, 1, 3, 2, 6])
F = ECDF(compl)
print(F(2), F(4), F(3) - F(1))                                  # shares
print(np.quantile(compl, [0.5, 0.8], method="inverted_cdf"))    # quantiles
plt.ecdf(compl)                                                 # step function
```

```
0.6 0.8 0.5
[2 3]
```

- `F(x)` is the share of observations $\leq x$ – also for unobserved values such as $x = 4$.
- `method="inverted_cdf"` is the course definition $x_{(\lceil n \cdot p \rceil)}$; the default interpolates.

▶ **Try it yourself:** notebook, section 1.1, exercise "Complaints per Day" – open in Colab

Overview

1 Empirical Cumulative Distribution Function

2 Histogram
- Exercises

3 Boxplot

4 Exercises

5 Python Commands

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Histogram

The histogram is a special type of bar chart used to visualise the distribution of metric data. What makes it special is that the observed values are not displayed individually but are grouped into **classes (bins/intervals)**.

Density

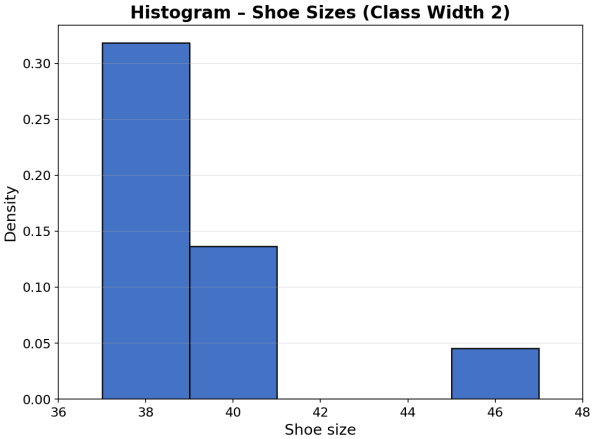
For each class, the **relative frequency** is computed and divided by the class width:

$$\text{density} = \frac{\text{relative frequency of the class}}{\text{class width}}$$

The bar height equals this density, so that the **area of each bar** represents the proportion of observations in the corresponding class.

- Classes are defined by interval boundaries
- The choice of class width has a strong effect on the shape of the histogram
- Classes of different widths are also possible

Histogram – Example



Histogram – Construction (Steps)

1. **Define the classes**
   - Notation: [= inclusive, (= exclusive
     - $[37, 39]$: both 37 and 39 belong to the class
     - $(39, 41]$: 39 does **not** belong, 41 does
   - Each class is open on the left and closed on the right: $(a, b]$
   - This way every value is assigned to **exactly one** class
   - Exception: the **first** class is closed on both sides $[a, b]$ so that the smallest value is not lost
   - e.g. $[37, 39]$, $(39, 41]$, $(41, 43]$, $(43, 45]$, $(45, 47]$
2. **Count the observations in each class** $\rightarrow$ absolute frequencies
3. **Compute the relative frequencies**
4. **Divide the relative frequencies by the class width** = density
5. **Draw the bars** with heights equal to these densities

Histogram – Construction (Table)

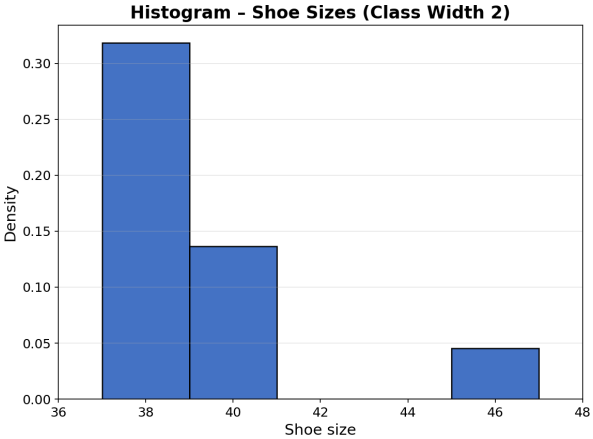
Shoe sizes: 40, 39, 38, 41, 46, 39, 39, 38, 37, 40, 39 $(n = 11)$

Class	$[37, 39]$	$(39, 41]$	$(41, 43]$	$(43, 45]$	$(45, 47]$	Total
Class width	2	2	2	2	2	
Abs. frequency	7	3	0	0	1	11
Rel. frequency	0.6364	0.2727	0	0	0.0909	1
Density $f_j/\Delta x_j$	0.3182	0.1364	0	0	0.0455	

Example: first class

Density $= \frac{0.6364}{2} = 0.3182$

Histogram – Example



Histogram – Alternative Classes (Class Width 3)

Shoe sizes: 40, 39, 38, 41, 46, 39, 39, 38, 37, 40, 39 $(n = 11)$

Class	$[37, 40]$	$(40, 43]$	$(43, 46]$	Total
Class width	3	3	3	
Abs. frequency	9	1	1	11
Rel. frequency	0.8182	0.0909	0.0909	1
Density $f_j/\Delta x_j$	0.2727	0.0303	0.0303	

Example: first class

Density $= \frac{0.8182}{3} = 0.2727$

Histogram – Unequal Class Widths

Shoe sizes: 40, 39, 38, 41, 46, 39, 39, 38, 37, 40, 39 $\quad (n = 11)$

Class	$[37, 39]$	$(39, 40]$	$(40, 41]$	$(41, 46]$	Total
Class width	2	1	1	5	
Abs. frequency	7	2	1	1	11
Rel. frequency	0.6364	0.1818	0.0909	0.0909	1
Density $f_j/\Delta x_j$	0.3182	0.1818	0.0909	0.0182	

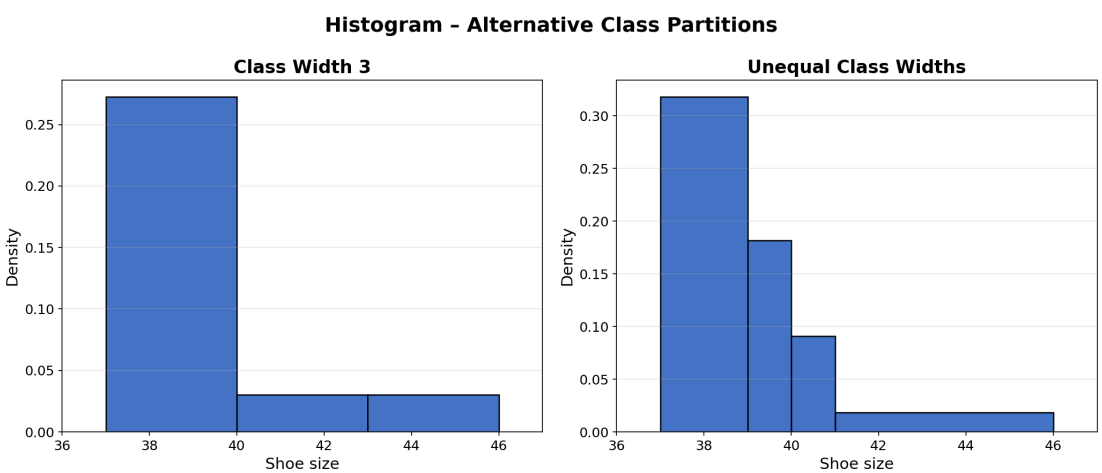
Example: last class

Density $= \frac{0.0909}{5} = 0.0182$

Key takeaway

The bar **height** is the density, the bar **area** is the share f_j: the wide class $(41, 46]$ holds as many players as $(40, 41]$, but its bar is only a fifth as high.

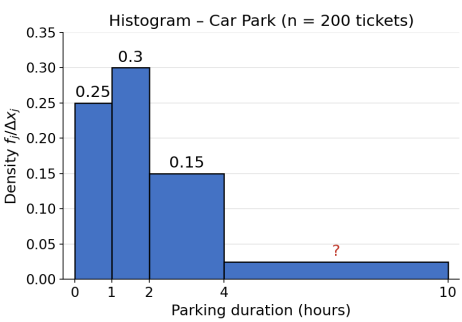
Histogram – Alternative Class Partition



Exercise – Car Park

Problem statement

Histogram of the parking durations (in hours) of $n = 200$ tickets of a car park. The bar labels are the densities; the last label is missing.



1. Determine the share f_j and the number h_j of tickets in each of the first three classes.
2. All bar areas add up to 1. Use this to determine f_4, h_4 and the missing bar height.
3. The bar over $(1, 2]$ is twice as high as the bar over $(2, 4]$. Does $(1, 2]$ contain twice as many tickets?
4. The first 2 hours are free of charge. What share of the tickets parked for free? How many tickets is that?

Solution (1/3) – Car Park: Shares from Areas

Rule: area = density · class width = f_j (share of the class); number $h_j = f_j \cdot n$. All areas together add up to 1.

Class	Δx_j	Density	f_j = density · Δx_j	$h_j = f_j \cdot 200$
$[0, 1]$	1	0.25	$0.25 \cdot 1 = \mathbf{0.25}$	**50**
$(1, 2]$	1	0.3	$0.3 \cdot 1 = \mathbf{0.3}$	**60**
$(2, 4]$	2	0.15	$0.15 \cdot 2 = \mathbf{0.3}$	**60**
$(4, 10]$	6	**0.025**	**0.15** (see below)	**30**
Total			1	200

2. Last class from "all areas add up to 1"

$f_4 = 1 - (0.25 + 0.3 + 0.3) = 1 - 0.85 = \mathbf{0.15} \quad \Rightarrow \quad h_4 = 0.15 \cdot 200 = \mathbf{30}$
missing height $= f_4/\Delta x_4 = 0.15/6 = \mathbf{0.025}$

Solution (2/3) – Car Park: Heights versus Areas

3. Twice as high $\neq$ twice as many

No. Both classes have the share 0.3, i.e. **60** tickets each. The class $(2, 4]$ is twice as wide (2 hours instead of 1), so the same area needs only half the height: $0.3/2 = 0.15$.

4. Free parking: at most 2 hours

Durations ≤ 2 hours = classes $[0, 1]$ and $(1, 2]$: $f_1 + f_2 = 0.25 + 0.3 = \mathbf{0.55}$, i.e. 55 % of the tickets $= 0.55 \cdot 200 = \mathbf{110}$ tickets. A ticket of exactly 2 hours lies in $(1, 2]$ and is therefore counted as free.

Key takeaway

Read a histogram by its **areas**, not by its heights: share = height $\cdot$ width. The height only says how densely the tickets lie per hour.

Solution (3/3) – Car Park: Python

Shares and numbers of tickets from the bar areas

```
w = np.diff([0, 1, 2, 4, 10])          # class widths: 1 1 2 6
dens = np.array([0.25, 0.3, 0.15])     # heights of the first three bars
f = dens * w[:3]                       # areas = shares: 0.25 0.3 0.3
f4 = 1 - f.sum()                       # all areas add up to 1: f4 = 0.15
print(round(f4 / w[3], 4), np.round(np.r_[f, f4] * 200), round(f[0] + f[1], 4))

0.025 [50. 60. 60. 30.] 0.55
```

- Output: missing bar height, tickets per class ($h_j = f_j \cdot 200$) and the share of free tickets (≤ 2 hours).
- Height $\cdot$ width $=$ area $=$ share f_j; the missing share is `1 - f.sum()`.

▶ **Try it yourself:** notebook, section 2, exercise "Car Park" – open in Colab

Histograms in Python

Class width 2

```
import pandas as pd
breaks = [37, 39, 41, 43, 45, 47]          # alternative: np.arange(37, 48, 2)
cls = pd.cut(shoe, breaks, right=True, include_lowest=True)  # [37,39], (39,41], ...
f = pd.Series(cls).value_counts(normalize=True, sort=False)   # rel. frequencies f_j
w = np.diff(breaks)                        # class widths
dens = f.to_numpy() / w                    # densities: 0.3182 0.1364 0 0 0.0455
plt.bar(breaks[:-1], dens, width=w, align="edge", edgecolor="black")
```

Unequal class widths (then the same lines as above)

```
breaks = [37, 39, 40, 41, 46]              # densities: 0.3182 0.1818 0.0909 0.0182
```

Why not simply `plt.hist`? It uses classes $[a, b)$ – see the next slide.

▶ **Try it yourself:** notebook, section 2 "Histogram" – open in Colab

Histograms in Python – Trap: Classes $[a, b)$

Trap: `plt.hist` and `np.histogram` use classes $[a, b)$

Ours: $(a, b]$, first class $[a, b]$. NumPy: a boundary value goes to the class on its **right**.

```
np.histogram(shoe, bins=[37, 39, 41, 43, 45, 47], density=True)[0]
# [0.1364 0.2727 0.0455 0. 0.0455]   <- not our densities!
```

Why?

39 (four times) and 41 lie on class boundaries: 39 falls into $[39, 41)$ instead of $[37, 39]$, 41 into $[41, 43)$ instead of $(39, 41]$ – `plt.hist` draws these wrong bars.

Remedy

Classes with `pd.cut(..., right=True, include_lowest=True)`, bars with `plt.bar` (next slide).

▶ **Try it yourself:** notebook, section 2 "Histogram" – open in Colab

Histograms in Python – with Our Classes

Correct: classes with `pd.cut`, bars with `plt.bar`

```
def density_hist(x, breaks):
    cls = pd.cut(x, breaks, right=True, include_lowest=True)
    f = pd.Series(cls).value_counts(normalize=True, sort=False)
    w = np.diff(breaks)                                    # class widths
    plt.bar(breaks[:-1], f / w, width=w, align="edge", edgecolor="black")
    return f / w                                           # densities f_j / w_j

density_hist(shoe, [37, 39, 41, 43, 45, 47])   # 0.3182 0.1364 0 0 0.0455
density_hist(shoe, [37, 39, 40, 41, 46])       # 0.3182 0.1818 0.0909 0.0182
```

- `right=True`: classes $(a, b]$; `include_lowest=True`: first class closed $[a, b]$.
- Bar heights = densities $f_j/\Delta x_j$ (as in the tables), so every bar area equals f_j.

▶ **Try it yourself:** notebook, section 2 "Histogram" – open in Colab

Classified Frequencies in Python

Class partition

```
breaks = [37, 39, 41, 43, 45, 47]          # class boundaries c0, c1, ..., ck
cls = pd.cut(shoe, breaks, right=True, include_lowest=True)
```

Absolute and relative classified frequencies

```
pd.Series(cls).value_counts(sort=False)                    # 7 3 0 0 1
pd.Series(cls).value_counts(normalize=True, sort=False)    # 0.6364 0.2727 0 0 0.0909
```

pandas labels the first class `(36.999, 39.0]` – this is exactly the closed class $[37, 39]$.

▶ **Try it yourself:** notebook, section 2 "Histogram" – open in Colab

Exercise – Histogram (Handball Team)

Problem statement

Shoe sizes of the handball team from Part 1 ($n = 13$):

$$40,\ 39,\ 38,\ 41,\ 46,\ 39,\ 39,\ 38,\ 37,\ 40,\ 39,\ 40,\ 42$$

1. Use the classes $[36, 38]$, $(38, 40]$, $(40, 42]$, $(42, 44]$, $(44, 46]$ (class width 2). Assign the values to the classes and compute the absolute and relative frequencies. Careful with values that lie exactly on a class boundary!
2. Compute the densities and sketch the histogram.
3. With the classes of the worked example, $[37, 39]$, $(39, 41]$, $\ldots$, $(45, 47]$, the same data give the counts $7, 4, 1, 0, 1$. Compare the shapes of the two histograms. What does this tell you about histograms?

Solution (1/2) – Histogram (Handball Team): Table

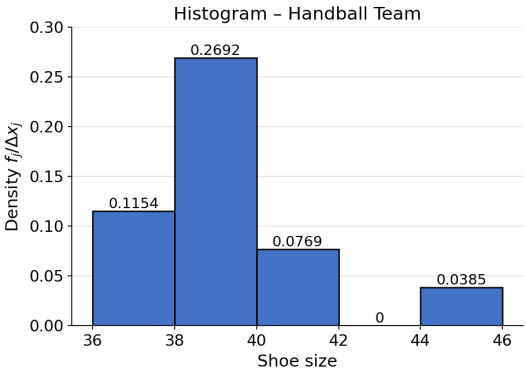
Rule: the first class is closed, all others are open on the left – a value on a boundary belongs to the class *below* it: $38 \to [36, 38]$, $40 \to (38, 40]$, $42 \to (40, 42]$, $46 \to (44, 46]$. Here 7 of the 13 values lie on a boundary.

Class	$[36, 38]$	$(38, 40]$	$(40, 42]$	$(42, 44]$	$(44, 46]$	Total
Values	37, 38, 38	4×39, 3×40	41, 42	–	46	
Class width Δx_j	2	2	2	2	2	
Abs. frequency h_j	3	7	2	0	1	13
Rel. frequency f_j	0.2308	0.5385	0.1538	0	0.0769	1
Density $f_j/\Delta x_j$	0.1154	0.2692	0.0769	0	0.0385	

Example: second class

$f_2 = 7/13 = 0.5385$, density $= 0.5385/2 = \mathbf{0.2692}$

Solution (2/2) – Histogram (Handball Team): Bars and Comparison



3. Same data, two impressions:

- $[37, 39], (39, 41], \ldots$: counts $7, 4, 1, 0, 1$ – the bars **fall from the first class on** ("the smallest sizes are the most common").
- $[36, 38], (38, 40], \ldots$: counts $3, 7, 2, 0, 1$ – the bars **rise to a peak** at $(38, 40]$ and then fall.

Shifting all boundaries by one size regroups the values: the four 39s now share a class with the three 40s instead of with 37 and 38.

Key takeaway

The shape of a histogram depends on the choice of classes: choose them deliberately and check whether a feature survives a different partition.

Exercise – Mobile Phone Costs

Problem statement

A survey of $n = 10$ students gives the following monthly mobile phone costs (in EUR):

$$12,\ 25,\ 6,\ 38,\ 15,\ 60,\ 9,\ 19,\ 32,\ 18$$

The values are to be grouped into the classes $[0, 10]$, $(10, 20]$, $(20, 40]$ and $(40, 80]$.

1. Determine the absolute and relative frequencies of the classes.
2. Compute the class widths and the densities (bar heights) and draw the histogram.
3. Why must the bar height *not* equal the relative frequency here? Which class is the most densely populated?
4. Which Python commands reproduce the frequencies and densities?

Solution (1/4) – Mobile Phone Costs: Class Frequencies

Sorted: 6, 9, 12, 15, 18, 19, 25, 32, 38, 60 $(n = 10)$

Class	Values in the class	h_j	$f_j = h_j/n$
$[0, 10]$	6, 9	2	0.2
$(10, 20]$	12, 15, 18, 19	4	0.4
$(20, 40]$	25, 32, 38	3	0.3
$(40, 80]$	60	1	0.1
Total		10	1

Assigning values to classes

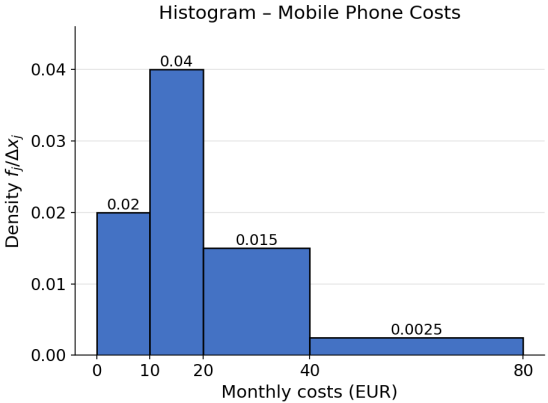
The first class is closed on both sides, all others are open on the left: a value of 10 would belong to $[0, 10]$, a value of 20 to $(10, 20]$.

Solution (2/4) – Mobile Phone Costs: Densities and Histogram

Rule: density $= f_j/\Delta x_j$ (class width) – this is the bar height, not the frequency.

Class	Δx_j	f_j	Density
$[0, 10]$	10	0.2	$0.2/10 = \mathbf{0.02}$
$(10, 20]$	10	0.4	$0.4/10 = \mathbf{0.04}$
$(20, 40]$	20	0.3	$0.3/20 = \mathbf{0.015}$
$(40, 80]$	40	0.1	$0.1/40 = \mathbf{0.0025}$

Check: area $=$ density $\cdot\, \Delta x_j = f_j$, e.g. $0.015 \cdot 20 = 0.3$; all areas add up to 1.



Solution (3/4) – Mobile Phone Costs: Interpretation

Why not the relative frequency as height?

In a histogram the **area** of a bar is the share of the class. With height f_j, the bar over $(20, 40]$ would have the area $0.3 \cdot 20 = 6$, the bar over $(10, 20]$ only $0.4 \cdot 10 = 4$ – the wide class with *fewer* observations would look bigger. Dividing by the class width makes classes of different widths comparable.

Most densely populated class

$(10, 20]$ with density **0.04**: per euro of class width, 4 % of the students lie in this class.

Solution (4/4) – Mobile Phone Costs: Python

Frequencies, densities and bars

```
costs = np.array([12, 25, 6, 38, 15, 60, 9, 19, 32, 18])
breaks = [0, 10, 20, 40, 80]
cls = pd.cut(costs, breaks, right=True, include_lowest=True)
h = pd.Series(cls).value_counts(sort=False).to_numpy()    # absolute frequencies
print(h, h / len(costs) / np.diff(breaks))                # h_j and densities
density_hist(costs, breaks)              # bars (function from "with Our Classes")
```

```
[2 4 3 1] [0.02   0.04   0.015  0.0025]
```

- `np.diff(breaks)` gives the unequal class widths $10, 10, 20, 40$.
- No value lies on a class boundary here, so `plt.hist` happens to give the same bars – `pd.cut` is right in every case.

▶ **Try it yourself:** notebook, section 2.1, exercise "Mobile Phone Costs" – open in Colab

Overview

1 Empirical Cumulative Distribution Function

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Boxplot

Guiding question

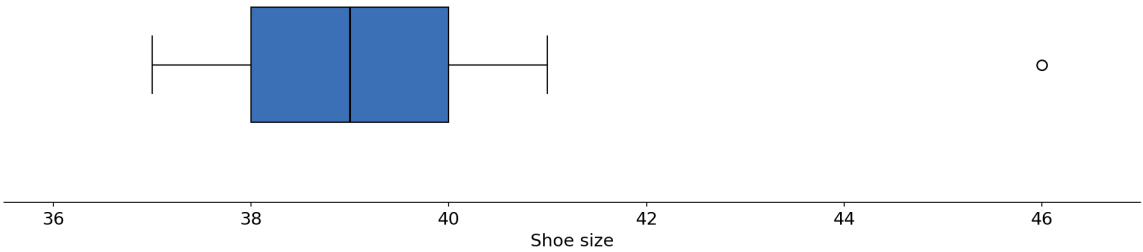
How can we see the centre, the spread, the skewness and the outliers of a dataset at one glance – and compare several groups side by side?

Definition

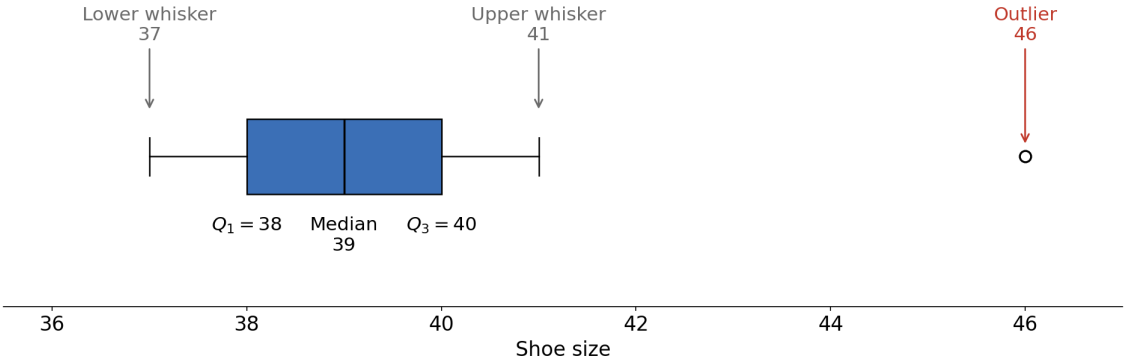
A graphical display that highlights only the most important summary values of a dataset and thereby visualises the distribution of the data.

- Building blocks: the quartiles Q_1, $\tilde{x}$, Q_3 of Part 1, two whiskers and the outliers
- Example: the shoe sizes of the soccer team (next slides)

Boxplot



Boxplot – with Labels



Boxplot – Construction

1. Sort the data
2. Determine the median
3. Determine the 1st quartile
4. Determine the 3rd quartile
5. Determine the lower whisker
6. Determine the upper whisker
7. Are there any outliers?
8. Draw the boxplot

Boxplot – Construction (Data)

Player	Shoe size	Sorted	
1	40	37	Lower whisker
2	39	38	
3	38	38	1st quartile
4	41	39	
5	46	39	
6	39	39	Median
7	39	39	
8	38	40	
9	37	40	3rd quartile
10	40	41	Upper whisker
11	39	46	Outlier
Sum	**436**	**436**	

Boxplot – Construction (Quartiles)

$n = 11$

	p	$n \cdot p$	$\lceil n \cdot p \rceil$ = position	Value
1st quartile	0.25	2.75	3	38
2nd quartile (median)	0.50	5.50	6	39
3rd quartile	0.75	8.25	9	40

Lower whisker	37
Q_1	38
Median	39
Q_3	40
Upper whisker	41
Outlier	46

Lower Whisker – Computation

The lower whisker is the **smallest data point that is not yet an outlier**. To find it, compute a lower bound – every value below it is classified as an outlier.

$$
\begin{aligned}
\text{Whisker}_{\text{min}} &= \min\{x_i \mid x_i \geq x_{0.25} - k(x_{0.75} - x_{0.25})\} \\
&\quad \text{with } k = 1.5 \\
&= \min\{x_i \mid x_i \geq 38 - 1.5(40 - 38)\} \\
&= \min\{x_i \mid x_i \geq 38 - 3\} \\
&= \min\{x_i \mid x_i \geq 35\} = 37
\end{aligned}
$$

Since 37 is the smallest data value ≥ 35, the lower whisker ends at 37.

Lower Whisker – Step by Step

1. **Compute the IQR:** the interquartile range is the difference between the 3rd and 1st quartile: $\text{IQR} = x_{0.75} - x_{0.25} = 40 - 38 = 2$
2. **Compute the lower bound:** subtract 1.5 times the IQR from the 1st quartile: $38 - 1.5 \cdot 2 = 38 - 3 = 35$
3. **Check the data:** every data point **below** 35 would be an outlier.
4. **Determine the lower whisker:** the smallest data value still ≥ 35 becomes the lower whisker. In our dataset this is 37.

Interpretation

The lower whisker extends down to 37 – the smallest value that is not classified as an outlier. If a value < 35 existed, it would be shown separately as an outlier in the boxplot.

Upper Whisker – Computation

The upper whisker is the **largest data point that is not yet an outlier**. To find it, compute an upper bound – every value above it is classified as an outlier.

$$\begin{aligned}\text{Whisker}_{\max} &= \max\{x_i \mid x_i \leq x_{0.75} + k(x_{0.75} - x_{0.25})\} \\ &\quad \text{with } k = 1.5 \\ &= \max\{x_i \mid x_i \leq 40 + 1.5(40 - 38)\} \\ &= \max\{x_i \mid x_i \leq 43\} = 41\end{aligned}$$

Since $46 > 43$, the value 46 is an outlier!

Upper Whisker – Step by Step

1. **Compute the IQR:** the interquartile range is the difference between the 3rd and 1st quartile: $\text{IQR} = x_{0.75} - x_{0.25} = 40 - 38 = 2$
2. **Compute the upper bound:** add 1.5 times the IQR to the 3rd quartile: $40 + 1.5 \cdot 2 = 40 + 3 = 43$
3. **Check the data:** every data point **above** 43 would be an outlier.
4. **Determine the upper whisker:** the largest data value still ≤ 43 becomes the upper whisker. In our dataset this is 41.

Interpretation

The upper whisker extends up to 41 – the largest value that is not classified as an outlier. Since $46 > 43$, the value 46 lies above the upper bound and is shown as a separate point in the boxplot.

Exercise – Daily Revenues

Problem statement

The online shop from the exercise "Complaints per Day" also records its daily revenue (in 1000 EUR) on the same $n = 10$ days:

Day i	1	2	3	4	5	6	7	8	9	10
Revenue x_i	15	18	12	22	17	13	14	20	17	4
Complaints	2	0	3	1	2	5	1	3	2	6

1. Sort the revenues and determine $x_{0.25}$, the median $\tilde{x}$ and $x_{0.75}$.
2. Compute the IQR and the bounds $x_{0.25} - 1.5 \cdot \text{IQR}$ and $x_{0.75} + 1.5 \cdot \text{IQR}$. Determine the lower and the upper whisker and possible outliers.
3. Draw the boxplot. Compare a possible outlier with the number of complaints on the same day.

Solution (1/2) – Daily Revenues: Quartiles and Bounds

Rule: position $= \lceil n \cdot p \rceil$ in the sorted data; if $n \cdot p$ is already an integer, it is **not** rounded up.

Sorted: 4, 12, **13**, 14, **15**, 17, 17, **18**, 20, 22 (positions 3, 5 and 8 in bold)

Quartiles ($n = 10$)

$x_{0.25}$: $10 \cdot 0.25 = 2.5 \rightarrow$ position 3 $\rightarrow$ **13**

$\tilde{x}$: $10 \cdot 0.5 = 5$ (integer) $\rightarrow$ position 5 $\rightarrow$ **15** (not the average of the 5th and 6th value)

$x_{0.75}$: $10 \cdot 0.75 = 7.5 \rightarrow$ position 8 $\rightarrow$ **18**

IQR and bounds ($k = 1.5$)

$\text{IQR} = x_{0.75} - x_{0.25} = 18 - 13 = \mathbf{5}$ $\quad 1.5 \cdot \text{IQR} = 1.5 \cdot 5 = 7.5$

lower bound: $13 - 7.5 = \mathbf{5.5}$ $\quad$ upper bound: $18 + 7.5 = \mathbf{25.5}$

Python check (`rev` = the ten revenues): `np.quantile(rev, [0.25, 0.5, 0.75], method="inverted_cdf")` $\rightarrow$ `[13 15 18]`, but `np.median(rev)` $\rightarrow$ `16.0` (average of the 5th and 6th value).

▶ **Try it yourself:** notebook, section 3, exercise "Daily Revenues" – open in Colab

Solution (2/2) – Daily Revenues: Whiskers, Outlier and Boxplot

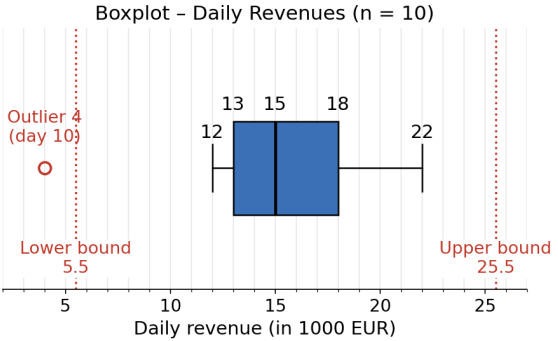
Rule: lower whisker = smallest value $\geq$ lower bound, upper whisker = largest value $\leq$ upper bound; all values outside the bounds are outliers.

Whiskers and outliers

$4 < 5.5 \rightarrow$ **4 is an outlier**
Lower whisker = smallest value ≥ 5.5: **12**
Upper whisker = largest value ≤ 25.5: **22**
(the maximum – no outlier at the top)

Outlier = day 10

On day 10 the shop also had the most complaints (6) – a hint at a disruption, e.g. a server outage. Check the cause; do not simply delete the value.



Unlike the shoe sizes, the outlier lies at the **bottom**: the lower whisker stops at 12, not at the minimum 4.

Boxplot – Interpretation (1/3): Symmetry and Median

Symmetry

A boxplot is **symmetric** when the median lies in the centre of the box and both whiskers have the same length. This indicates that the data are evenly distributed around the centre.

Position of the median

The median splits the box into two halves. If it does **not** lie in the middle, the distribution is **skewed**:

- Median in the **left** half of the box $\rightarrow$ **right-skewed** (positively skewed): many small values, a few large values pull the distribution to the right.
- Median in the **right** half of the box $\rightarrow$ **left-skewed** (negatively skewed): many large values, a few small values pull the distribution to the left.

Boxplot – Interpretation (2/3): Box and Whiskers

Length of the box (IQR)

The box contains the middle 50 % of the data – from the 1st quartile (Q_1) to the 3rd quartile (Q_3). A **wide box** indicates that these middle values are widely spread. A **narrow box** shows that half of the data lie close together.

Length of the whiskers

The whiskers extend from the box to the smallest and largest value that is not an outlier. If both whiskers are **equally long**, the distribution is symmetric in the tails. **Different whisker lengths** indicate that the data are more spread out in one direction than in the other.

Boxplot – Interpretation (3/3): Outliers

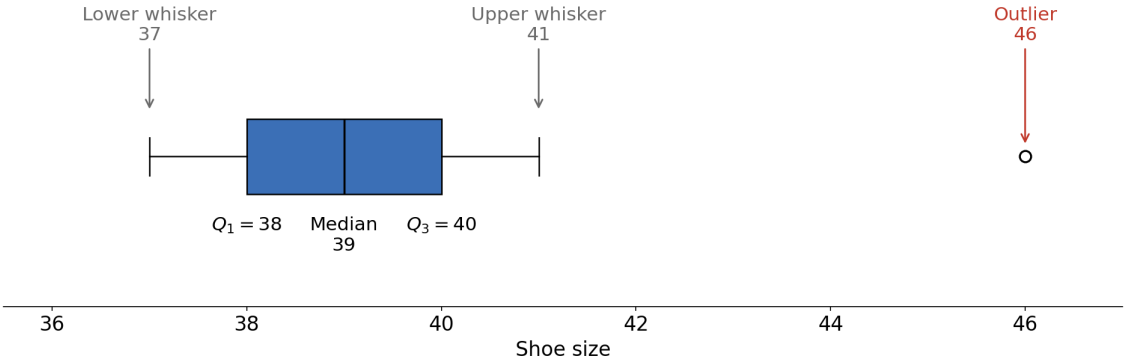
Outliers

Values that lie beyond the whiskers are shown individually as **outliers**. A value is considered an outlier if it lies more than 1.5 times the IQR below Q_1 or above Q_3. Outliers can indicate measurement errors, special cases, or a strongly skewed distribution.

Example: shoe-size data

- Box: $Q_1 = 38$, median $= 39$, $Q_3 = 40$ $\rightarrow$ the median lies in the centre, the box is symmetric.
- Whiskers: lower whisker $= 37$, upper whisker $= 41$ $\rightarrow$ both whiskers have the same length (each 1 unit from the box).
- Outlier: $46 > 43$ (upper bound) $\rightarrow$ despite the symmetric box, the overall boxplot is **not** symmetric.

Boxplot – with Labels



Boxplot – Conclusion

Advantages of the boxplot

- Compact display of the **most important summary statistics** at a glance: median, quartiles, range and outliers.
- Allows a quick assessment of the **symmetry** and **skewness** of a distribution.
- Especially useful for **comparing several groups** side by side – e.g. shoe sizes of different teams.
- **Outliers** are automatically detected and made visible.

Limitation

The boxplot reveals no information about the exact shape of the distribution inside the box – two very different datasets can produce the same boxplot. To study the shape, a histogram is a useful complement.

Exercise – Processing Times of Two Teams

Problem statement

Two service teams handle customer requests. For $n = 11$ requests each, the processing time (in minutes) was recorded:

Team A	7, 4, 9, 6, 18, 8, 5, 10, 6, 9, 7
Team B	12, 8, 15, 5, 14, 10, 16, 8, 11, 14, 7

1. Team A: determine the quartiles, the IQR, the bounds $x_{0.25} - 1.5 \cdot \text{IQR}$ and $x_{0.75} + 1.5 \cdot \text{IQR}$, the whiskers and possible outliers.
2. Carry out the same computations for team B.
3. Draw both boxplots above a common axis.
4. Compare median, IQR and outliers. What does the position of the median in the box say about skewness? Which team works faster and more consistently?

Solution (1/5) – Processing Times: Team A

Sorted: 4, 5, **6**, 6, 7, **7**, 8, 9, **9**, 10, 18 $\quad (n = 11)$

Quartiles (position $\lceil n \cdot p \rceil$ in the sorted data)

$x_{0.25}$: $11 \cdot 0.25 = 2.75 \rightarrow$ position 3 $\rightarrow$ **6**
$\tilde{x}$: $11 \cdot 0.5 = 5.5 \rightarrow$ position 6 $\rightarrow$ **7**
$x_{0.75}$: $11 \cdot 0.75 = 8.25 \rightarrow$ position 9 $\rightarrow$ **9**

IQR and bounds ($k = 1.5$)

IQR $= 9 - 6 =$ **3** lower bound: $6 - 1.5 \cdot 3 =$ **1.5** upper bound: $9 + 1.5 \cdot 3 =$ **13.5**

Whiskers and outliers

Lower whisker = smallest value ≥ 1.5: **4** Upper whisker = largest value ≤ 13.5: **10**
$18 > 13.5 \rightarrow$ **18 is an outlier** (one unusually long request).

Solution (2/5) – Processing Times: Team B

Sorted: 5, 7, **8**, 8, 10, **11**, 12, 14, **14**, 15, 16 $(n = 11)$

Quartiles (same positions as for team A: 3, 6, 9)

$x_{0.25} = \mathbf{8}$ $\tilde{x} = \mathbf{11}$ $x_{0.75} = \mathbf{14}$

IQR and bounds ($k = 1.5$)

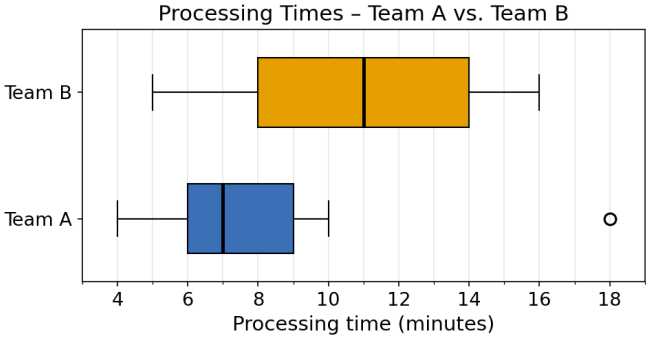
IQR $= 14 - 8 = \mathbf{6}$ lower bound: $8 - 1.5 \cdot 6 = \mathbf{-1}$ upper bound: $14 + 1.5 \cdot 6 = \mathbf{23}$

Whiskers and outliers

Lower whisker = smallest value ≥ -1: **5** Upper whisker = largest value ≤ 23: **16**
All values lie within the bounds $\rightarrow$ **no outlier**; the whiskers reach the minimum and the maximum.

Solution (3/5) – Processing Times: Boxplots

	Team A	Team B
Lower whisker	4	5
$x_{0.25}$	6	8
Median	7	11
$x_{0.75}$	9	14
Upper whisker	10	16
Outliers	18	–
IQR	3	6



Common axis: box from $x_{0.25}$ to $x_{0.75}$, line at the median, whiskers to the lower and upper whisker, outliers as single points.

Solution (4/5) – Processing Times: Comparison

Comparing the two boxplots

- **Median:** team B (11) $>$ team A (7) – team A typically handles requests faster.
- **IQR:** team B (6) $>$ team A (3) – the middle 50 % of B's times are spread twice as widely; team A works more consistently.
- **Outliers:** only team A (18 minutes) – a single case, e.g. a particularly complex request.
- **Skewness:** at A the median (7) lies in the left half of the box $[6, 9]$ ($7 - 6 = 1 < 9 - 7 = 2$) and the outlier lies far to the right $\rightarrow$ right-skewed (the whiskers alone, 2 vs. 1 minute, would not show this). At B the median (11) lies exactly in the middle of $[8, 14]$, the whiskers are 3 and 2 minutes long $\rightarrow$ roughly symmetric.

Conclusion

Team A is faster and more consistent (apart from one outlier); team B is slower and its processing times spread more.

Solution (5/5) – Processing Times: Python

Boxplot statistics with the course quartiles, then bxp

```
def box_stats(x, k=1.5):                          # course quartiles + 1.5 IQR rule
    q1, med, q3 = np.quantile(x, [0.25, 0.5, 0.75], method="inverted_cdf")
    lo, hi = q1 - k * (q3 - q1), q3 + k * (q3 - q1)
    inside = x[(x >= lo) & (x <= hi)]
    return dict(q1=q1, med=med, q3=q3, whislo=inside.min(), whishi=inside.max(),
                fliers=x[(x < lo) | (x > hi)])
team_a = np.array([7, 4, 9, 6, 18, 8, 5, 10, 6, 9, 7])
team_b = np.array([12, 8, 15, 5, 14, 10, 16, 8, 11, 14, 7])
stats = [box_stats(team_a), box_stats(team_b)]
# whislo q1 med q3 whishi fliers   A: 4 6 7 9 10 [18]   B: 5 8 11 14 16 []
fig, ax = plt.subplots(); ax.bxp(stats, vert=False)   # boxplots, common axis
ax.set_yticks([1, 2], ["Team A", "Team B"])
```

`plt.boxplot` interpolates the quartiles – for these data it happens to give the same result, but not in general.

▶ **Try it yourself:** notebook, section 3.1, exercise "Processing Times of Two Teams" – open in Colab

Overview

1 Empirical Cumulative Distribution Function

2 Histogram

3 Boxplot

4 Exercises
- Body Heights
- Café Chain (Exam Style)

5 Python Commands

6 Summary

Exercise: Body Heights – Problem Statement

The body heights (in cm) of $n = 15$ persons are studied:

164, 162, 168, 198, 166, 166, 192, 160, 151, 175, 180, 160, 168, 175, 168

Sorted:

151, 160, 160, 162, 164, 166, 166, 168, 168, 168, 175, 175, 180, 192, 198

1. Compute the mean, median, quartiles and the 17 % quantile.
2. Compute the measures of dispersion: range, IQR, sample variance, standard deviation, coefficient of variation and normalised coefficient of variation.
3. Draw the empirical CDF, a histogram (class width 10 cm) and a boxplot.
4. Draw the empirical CDF for the classified data (class midpoints).

Exercise: Body Heights – The Data

Person	Height	Sorted	Abs. freq.	Rel. freq.	Cum. freq.
1	164	151	1	$0.0\overline{6}$	$0.0\overline{6}$
2	162	160	2	$0.1\overline{3}$	0.2
3	168	160			
4	198	162	1	$0.0\overline{6}$	$0.2\overline{6}$
5	166	164	1	$0.0\overline{6}$	$0.3\overline{3}$
6	166	166	2	$0.1\overline{3}$	$0.4\overline{6}$
7	192	166			
8	160	168	3	0.2	$0.\overline{6}$
9	151	168			
10	175	168			
11	180	175	2	$0.1\overline{3}$	0.8
12	160	175			
13	168	180	1	$0.0\overline{6}$	$0.8\overline{6}$
14	175	192	1	$0.0\overline{6}$	$0.9\overline{3}$
15	168	198	1	$0.0\overline{6}$	1
Sum	**2553**	**2553**	**15**	**1**	

Note: $\overline{6}$ denotes a repeating decimal, e.g.

$$0.0\overline{6} = 0.0666\ldots = \tfrac{1}{15}$$

Exercise: Body Heights – Results

Statistic	Value	Bound
Sum	2553	
Arithmetic mean	170.2	
Median	168	
Mode	168	
1st quartile	162	
3rd quartile	175	
17 % quantile	160	
Lower whisker	151	142.5
Upper whisker	192	194.5
Sample variance	151.60	
Standard deviation	12.3126	
Coefficient of variation	0.0723	
Normalised coeff. of variation	0.0187	

Exercise: Body Heights – Mean (Step by Step)

Formula

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

Step 1: Sum all values:

$$\begin{aligned} \sum x_i &= 164 + 162 + 168 + 198 + 166 + 166 + 192 \\ &\quad + 160 + 151 + 175 + 180 + 160 + 168 + 175 + 168 \\ &= 2553 \end{aligned}$$

Step 2: Divide by $n = 15$:

$$\bar{x} = \frac{2553}{15} = 170.2\,\text{cm}$$

Exercise: Body Heights – Median and Mode

Sorted data:

151, 160, 160, 162, 164, 166, 166, **168**, 168, 168, 175, 175, 180, 192, 198

Median ($p = 0.5$)

$n \cdot p = 15 \cdot 0.5 = 7.5$ – not an integer $\rightarrow$ round up to position 8
The 8th value in the sorted sequence is the median: $\tilde{x} = x_{0.5} = \mathbf{168}$ cm.

Mode

The most frequent value: 168 occurs 3 times $\rightarrow$ **168** cm.

Exercise: Body Heights – Quartiles and Quantile

Rule: position $= \lceil n \cdot p \rceil$ – compute $n \cdot p$ and round up (if $n \cdot p$ is an integer, it is already the position).

17 % quantile ($p = 0.17$)

$n \cdot p = 15 \cdot 0.17 = 2.55 \rightarrow$ round up to position 3 $\rightarrow x_{0.17} = 160\,\text{cm}$

1st quartile ($p = 0.25$)

$n \cdot p = 15 \cdot 0.25 = 3.75 \rightarrow$ round up to position 4 $\rightarrow x_{0.25} = 162\,\text{cm}$

3rd quartile ($p = 0.75$)

$n \cdot p = 15 \cdot 0.75 = 11.25 \rightarrow$ round up to position 12 $\rightarrow x_{0.75} = 175\,\text{cm}$

Exercise: Body Heights – Range and IQR

Range

$R = x_{\text{max}} - x_{\text{min}} = 198 - 151 = \mathbf{47}\,\text{cm}$

Interquartile range (IQR)

$\text{IQR} = x_{0.75} - x_{0.25} = 175 - 162 = \mathbf{13}\,\text{cm}$

The range reacts strongly to outliers (here: 198), whereas the IQR is unaffected by them because it only considers the middle 50 % of the data.

Exercise: Body Heights – Variance and Standard Deviation

Sample variance

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

Step 1: Square the deviations from the mean:

$$(151 - 170.2)^2 = 368.64, \quad (160 - 170.2)^2 = 104.04, \quad \ldots$$

Step 2: Sum of squared deviations: $\sum(x_i - \bar{x})^2 = 2122.40$

Step 3: Divide by $n - 1 = 14$: $s^2 = \frac{2122.40}{14} = \mathbf{151.60}\,\text{cm}^2$

Standard deviation

$s = \sqrt{s^2} = \sqrt{151.60} = \mathbf{12.3126}\,\text{cm}$

Exercise: Body Heights – Coefficient of Variation

Coefficient of variation

$$v = \frac{s}{\bar{x}} = \frac{12.3126}{170.2} = \mathbf{0.0723}$$

The coefficient of variation expresses the dispersion relative to the mean – here the values scatter by about 7.2 % of the mean.

Normalised coefficient of variation

$$v^* = \frac{v}{\sqrt{n}} = \frac{0.0723}{\sqrt{15}} = \frac{0.0723}{3.8730} = \mathbf{0.0187}$$

The normalised coefficient of variation additionally takes the sample size into account: v is divided by $\sqrt{n}$, its largest possible value for non-negative data, so that $0 \leq v^* \leq 1$.

Exercise: Body Heights – Whisker Computation

Lower bound

$x_{0.25} - 1.5 \cdot \text{IQR} = 162 - 1.5 \cdot 13 = 162 - 19.5 = \mathbf{142.5}$

Smallest value ≥ 142.5: lower whisker $= 151$

Upper bound

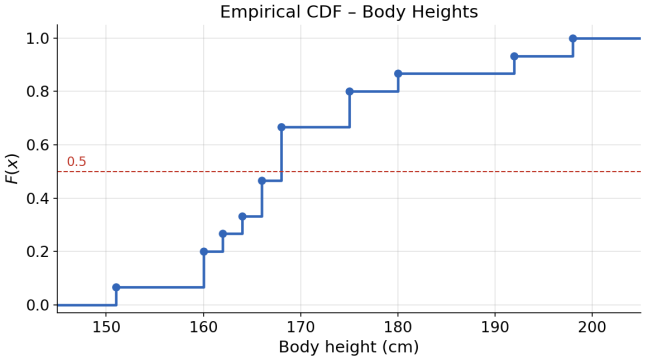
$x_{0.75} + 1.5 \cdot \text{IQR} = 175 + 1.5 \cdot 13 = 175 + 19.5 = \mathbf{194.5}$

Largest value ≤ 194.5: upper whisker $= 192$

Outlier

$198 > 194.5 \rightarrow$ the value 198 lies above the upper bound and is therefore an outlier.

Exercise: Body Heights – Empirical CDF



Step function from the column "Cum. freq." of the data table: at every observed value it jumps up by f_j (e.g. by 0.2 at 168 cm, which occurs 3 times).

Reading off quantiles

$x_p = \min\{x \mid F(x) \geq p\}$

- $x_{0.17}$: $F(160) = 0.2 \geq 0.17 \rightarrow 160$
- $x_{0.25}$: $F(162) = 0.2\overline{6} \geq 0.25 \rightarrow 162$
- $\tilde{x}$: $F(168) = 0.\overline{6} \geq 0.5 \rightarrow 168$
- $x_{0.75}$: $F(175) = 0.8 \geq 0.75 \rightarrow 175$

Same results as with the position rule.

Exercise: Body Heights – Histogram (Table)

Class width 10 cm, first class closed on both sides $(n = 15)$

Class	$[150, 160]$	$(160, 170]$	$(170, 180]$	$(180, 190]$	$(190, 200]$	Total
Class width	10	10	10	10	10	
Abs. frequency	3	7	3	0	2	15
Rel. frequency	0.2	$0.4\overline{6}$	0.2	0	$0.1\overline{3}$	1
Density $f_j/\Delta x_j$	0.02	0.0467	0.02	0	0.0133	

Example: second class $(160, 170]$

Values 162, 164, 166, 166, 168, 168, 168 $\rightarrow h_2 = 7$, $f_2 = \frac{7}{15} = 0.4\overline{6}$, density $= \frac{0.4\overline{6}}{10} = 0.0467$

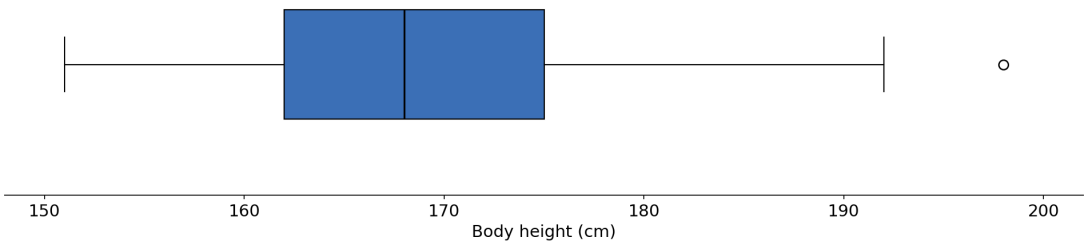
Boundary values

160 (twice) belongs to $[150, 160]$ and 180 to $(170, 180]$ – classes are closed on the **right**.

Exercise: Body Heights – Histogram (Class Width 10 cm)



Exercise: Body Heights – Boxplot



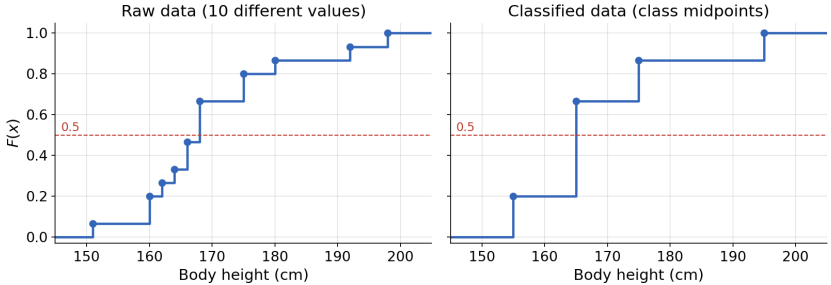
Exercise: Body Heights – Classified Data (Class Width 10 cm)

Class	$[150, 160]$	$(160, 170]$	$(170, 180]$	$(180, 190]$	$(190, 200]$	Total
Class midpoint a_j	155	165	175	185	195	
Abs. frequency h_j	3	7	3	0	2	15
Rel. frequency f_j	0.2	$0.4\overline{6}$	0.2	0	$0.1\overline{3}$	1
Cum. frequency $F(a_j)$	0.2	$0.\overline{6}$	$0.8\overline{6}$	$0.8\overline{6}$	1	

Construction

All observations of a class are placed at its **class midpoint** a_j; the empirical CDF then jumps by f_j at each midpoint.

Exercise: Body Heights – Empirical CDF of the Classified Data



- Only 4 steps instead of 10 – the classified CDF is much **coarser**, but its overall shape is similar.
- It reaches 0.5 at $a_2 = 165$, whereas the exact median is 168 cm.

Interpretation

Classifying the data simplifies the picture but **loses information** within the classes.

Exercise: Body Heights – Python

```
heights = np.array([164, 162, 168, 198, 166, 166, 192, 160, 151, 175,
                    180, 160, 168, 175, 168])
np.quantile(heights, [0.17, 0.25, 0.5, 0.75], method="inverted_cdf")  # 160 162 168 175
breaks = [150, 160, 170, 180, 190, 200]
cls = pd.cut(heights, breaks, right=True, include_lowest=True)
h = pd.Series(cls).value_counts(sort=False).to_numpy()    # 3 7 3 0 2
density_hist(heights, breaks)                             # 0.02 0.0467 0.02 0 0.0133
plt.figure(); plt.ecdf(heights)                           # new figure: raw-data ECDF
classified = np.repeat([155, 165, 175, 185, 195], h)      # each value -> class midpoint
plt.ecdf(classified)                                      # classified ECDF, same axes
np.quantile(classified, 0.5, method="inverted_cdf")       # 165 (exact median: 168)
```

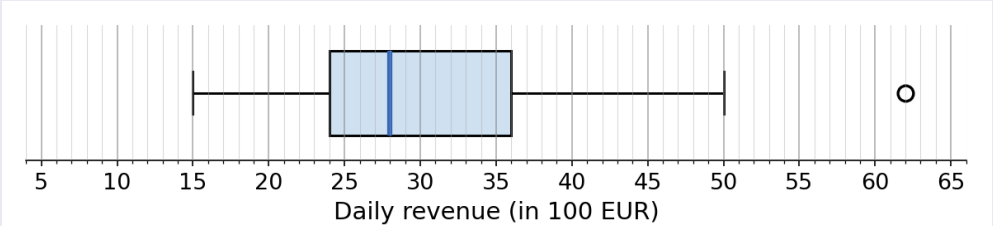
Trap again: `plt.hist(heights, bins=breaks)`

Classes $[a, b)$ put 160 and 180 into the next class: counts $1, 9, 2, 1, 2$ instead of $3, 7, 3, 0, 2$.

▶ **Try it yourself:** notebook, section 4.1 "Body Heights" – open in Colab

Exercise – Café Chain (Exam Style): Reading a Boxplot

Problem statement (Part A): boxplot of the $n = 60$ daily revenues of a café

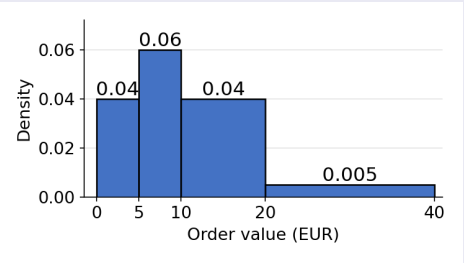


1. Read off the median, Q_1, Q_3 and the whisker ends; compute the IQR.
2. On at least how many days was the revenue at most 2800 EUR?
3. Compute the fences. Why is 62 drawn as a point, and why does the upper whisker end at 50?
4. Assess the skewness. Is the mean above or below the median?

Exercise – Café Chain (Exam Style): Histogram and Robust Measures

Problem statement (Parts B and C)

B: Histogram of the values (in EUR) of $n = 50$ orders; the bar labels are the densities.



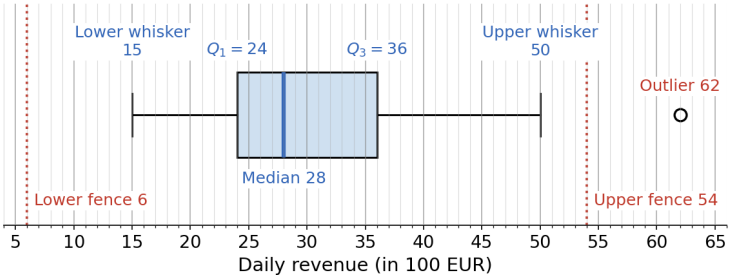
5. Determine the relative frequency and the number of orders of every class.
6. Which class contains the most orders? Is it the class with the highest bar?
7. What share of the orders is worth more than 10 EUR?

C: Repair times of the espresso machine (minutes): 2, 3, 3, 4, 5, 6, 26.

8. Compute the mean, the median and the trimmed mean (smallest and largest value removed). Is 26 an outlier by the $1.5 \cdot \text{IQR}$ rule? Which measure would you report?

Solution (1/5) – Café Chain: Reading Off the Boxplot

Rule: box edges = Q_1 and Q_3, line in the box = median, whisker ends = the most extreme *data values* inside the fences.



1. Median = **28** (2800 EUR), $Q_1 = \mathbf{24}$, $Q_3 = \mathbf{36}$, whiskers at **15** and **50**;
   IQR $= 36 - 24 = \mathbf{12}$: the middle 50 % of the days lie within a range of 1200 EUR.

Solution (2/5) – Café Chain: Share of Days and Fences

2. **Rule:** median $= x_{(\lceil n \cdot 0.5 \rceil)}$ (Part 1).

$$\tilde{x} = x_{(\lceil 60 \cdot 0.5 \rceil)} = x_{(30)} = 28$$

The 30 smallest revenues are all $\leq 28 \rightarrow$ on **at least 30 days** (half of the days) the revenue was at most 2800 EUR, i.e. $F(28) \geq 0.5$. (Ties may make it more days.)

3. **Rule:** fences $Q_1 - 1.5 \cdot \text{IQR}$ and $Q_3 + 1.5 \cdot \text{IQR}$.

$$24 - 1.5 \cdot 12 = 24 - 18 = \mathbf{6}, \qquad 36 + 1.5 \cdot 12 = 36 + 18 = \mathbf{54}$$

- $62 > 54 \rightarrow$ **outlier**, drawn as a single point.
- The upper whisker ends at the largest *observed* value ≤ 54, which is 50 – the fence 54 itself is no data value.
- $15 \geq 6$: no outlier at the bottom; the lower whisker is the minimum.

Solution (3/5) – Café Chain: Skewness

4. Three indications point in the same direction:
   - median $- Q_1 = 28 - 24 = 4 <$ $Q_3 -$ median $= 36 - 28 = 8$: the median lies in the left half of the box,
   - lower whisker $24 - 15 = 9 <$ upper whisker $50 - 36 = 14$,
   - the only outlier (62) lies at the top.

   $\rightarrow$ **right-skewed**: a few days with very high revenue stretch the right tail. They pull the mean upwards, so we expect $\bar{x} > \tilde{x} = 28$, i.e. the mean **above** the median.

Key takeaway

Reading a boxplot: the box edges are quartiles, the whisker ends are observed values (not the fences). Skewness shows in the position of the median in the box, the whisker lengths and the side of the outliers.

Solution (4/5) – Café Chain: Reading the Histogram

5. Rule: area = density · class width = f_j (relative frequency); number of orders $h_j = f_j \cdot n$.

Class	Δx_j	Density	$f_j = \text{density} \cdot \Delta x_j$	$h_j = f_j \cdot 50$
$[0, 5]$	5	0.04	0.2	10
$(5, 10]$	5	0.06	0.3	15
$(10, 20]$	10	0.04	0.4	20
$(20, 40]$	20	0.005	0.1	5
Total			1	50

6. Most orders: $(\mathbf{10}, \mathbf{20}]$ with 20 orders (40 %) – although its bar (0.04) is *lower* than that of $(5, 10]$ (0.06): it is twice as wide. The highest bar shows where the orders lie most *densely* (per euro), not where most of them lie.
7. More than 10 EUR: $f_3 + f_4 = 0.4 + 0.1 = \mathbf{0.5}$, i.e. 50 % of the orders (25 orders).

Solution (5/5) – Café Chain: Robust Measures of Location

Sorted: 2, 3, 3, 4, 5, 6, 26 $\quad (n = 7)$

8. **Mean** $\bar{x} = 49/7 = \mathbf{7}$; **median** $= x_{(\lceil 3.5 \rceil)} = x_{(4)} = \mathbf{4}$; **trimmed mean** $= (3 + 3 + 4 + 5 + 6)/5 = 21/5 = \mathbf{4.2}$.

   Outlier check: $Q_1 = x_{(\lceil 1.75 \rceil)} = x_{(2)} = 3$, $Q_3 = x_{(\lceil 5.25 \rceil)} = x_{(6)} = 6$, IQR $= 3$, upper fence $6 + 4.5 = 10.5 \rightarrow 26$ is an **outlier**.

Which measure to report?

The mean (7) exceeds six of the seven values – the outlier pulls it up. Median (4) and trimmed mean (4.2) are robust $\rightarrow$ report the **median** (or the trimmed mean).

```
rep = np.array([2, 3, 3, 4, 5, 6, 26])
print(rep.mean(), np.quantile(rep, 0.5, method="inverted_cdf"), np.sort(rep)[1:-1].mean())
```

```
7.0 4 4.2
```

▶ **Try it yourself:** notebook, section 4.2 "Café Chain (Exam Style)" – open in Colab

Overview

1 Empirical Cumulative Distribution Function

2 Histogram

3 Boxplot

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5 Python Commands

6 Summary

Python Commands: Frequencies and Measures of Central Tendency

Absolute frequencies

```
import numpy as np, pandas as pd
shoe = np.array([40, 39, 38, 41, 46, 39, 39, 38, 37, 40, 39])
tab = pd.Series(shoe).value_counts().sort_index()    # sorted by shoe size
print(tab)                                           # 37:1 38:2 39:4 40:2 41:1 46:1
```

Mean, median and quantiles

```
np.mean(shoe)      # arithmetic mean: 39.6364
np.median(shoe)    # 39.0 -- note: averages for even n, not our definition!
# Better: method="inverted_cdf" matches our definition x_(ceil(n*p)):
np.quantile(shoe, 0.8, method="inverted_cdf")                  # 80% quantile: 40
np.quantile(shoe, [0.25, 0.5, 0.75], method="inverted_cdf")  # quartiles: 38 39 40
```

▶ **Try it yourself:** notebook, section 5 "Python Commands" – open in Colab

Python Commands: Measures of Dispersion

Sample variance and standard deviation

```
np.var(shoe, ddof=1)     # s^2 = 5.6545   (ddof=1: divide by n - 1)
np.std(shoe, ddof=1)     # s   = 2.3779
```

Coefficient of variation and normalised coefficient of variation

```
v = np.std(shoe, ddof=1) / np.mean(shoe)    # 0.0600
vstar = v / np.sqrt(len(shoe))              # 0.0181
```

Without `ddof=1`, NumPy divides by n (5.1405 instead of $s^2 = 5.6545$); pandas' `.var()` and `.std()` already use `ddof=1`.

▶ **Try it yourself:** notebook, section 5 "Python Commands" – open in Colab

Python Commands: Empirical CDF

Computing and evaluating: shares and quantiles

```
from statsmodels.distributions.empirical_distribution import ECDF
F = ECDF(shoe)
F(39)            # value of the ECDF at x = 39: 0.6364
F(40) - F(38)    # share in (38, 40], i.e. sizes 39 and 40: 0.5455 (= 6/11)
np.quantile(shoe, [0.5, 0.8], method="inverted_cdf")    # min{x | F(x) >= p}: 39 40
```

Plotting

```
import matplotlib.pyplot as plt
plt.ecdf(shoe)  # step function with vertical jumps
plt.show()
```

▶ **Try it yourself:** notebook, section 5 "Python Commands" – open in Colab

Python Commands: Histogram

Equal class widths (here: 2)

```
density_hist(shoe, [37, 39, 41, 43, 45, 47])   # 0.3182 0.1364 0 0 0.0455
density_hist(shoe, np.arange(37, 48, 2))       # same breaks, generated
```

Unequal class widths

```
density_hist(shoe, [37, 39, 40, 41, 46])       # 0.3182 0.1818 0.0909 0.0182
```

`density_hist` (`pd.cut` + `plt.bar`) is defined in the Histogram section. Do **not** use `plt.hist(shoe, bins=breaks, density=True)`: its classes are $[a, b)$, so 39 and 41 would be counted in the wrong class.

▶ **Try it yourself:** notebook, section 5 "Python Commands" – open in Colab

Python Commands: Classified Frequencies

Absolute frequencies

```
pd.Series(shoe).value_counts().sort_index()
```

Class partition

```
cls = pd.cut(shoe, bins=[37, 39, 41, 43, 45, 47], right=True, include_lowest=True)
```

Absolute and relative classified frequencies

```
pd.Series(cls).value_counts(sort=False)                    # 7 3 0 0 1
pd.Series(cls).value_counts(normalize=True, sort=False)    # 0.6364 0.2727 0 0 0.0909
```

▶ **Try it yourself:** notebook, section 5 "Python Commands" – open in Colab

Python Commands: Boxplot

Standard boxplot (Tukey)

```
plt.boxplot(shoe, vert=False, whis=1.5)   # horizontal, whiskers at k = 1.5
```

`whis` can be omitted for the default $k = 1.5$.

Trap: matplotlib interpolates the quartiles

`plt.boxplot` uses NumPy's default (linear interpolation), not our definition $x_{(\lceil n \cdot p \rceil)}$:

```
np.quantile(shoe, [0.25, 0.75])                          # [38.5 40.] interpolated
np.quantile(shoe, [0.25, 0.75], method="inverted_cdf")   # [38 40]    our definition
```

The box would start at 38.5 instead of $Q_1 = 38$ – next slide: boxplot with our quartiles.

▶ **Try it yourself:** notebook, section 5 "Python Commands" – open in Colab

Python Commands: Boxplot with Our Quartile Definition

Quartiles, whisker bounds, whiskers and outliers

```
q1, med, q3 = np.quantile(shoe, [0.25, 0.5, 0.75], method="inverted_cdf")  # 38 39 40
iqr = q3 - q1                                                              # IQR = 2
lo, hi = q1 - 1.5 * iqr, q3 + 1.5 * iqr                                    # bounds: 35.0  43.0
inside = shoe[(shoe >= lo) & (shoe <= hi)]
whislo, whishi = inside.min(), inside.max()                                # whiskers: 37  41
outliers = shoe[(shoe < lo) | (shoe > hi)]                                 # outlier: [46]
```

Drawing the boxplot from exactly these statistics

```
stats = dict(q1=q1, med=med, q3=q3, whislo=whislo, whishi=whishi, fliers=outliers)
plt.gca().bxp([stats], vert=False)
plt.show()
```

▶ **Try it yourself:** notebook, section 5 "Python Commands" – open in Colab

Python Commands: Boxplots of Several Groups

Boxplot statistics with our quartiles – for any number of groups

```
def box_stats(x, k=1.5):
    q1, med, q3 = np.quantile(x, [0.25, 0.5, 0.75], method="inverted_cdf")
    lo, hi = q1 - k * (q3 - q1), q3 + k * (q3 - q1)      # fences
    inside = x[(x >= lo) & (x <= hi)]
    return dict(q1=q1, med=med, q3=q3, whislo=inside.min(), whishi=inside.max(),
                fliers=x[(x < lo) | (x > hi)])

fig, ax = plt.subplots()                          # several boxplots, common axis
ax.bxp([box_stats(team_a), box_stats(team_b)], vert=False)
ax.set_yticks([1, 2], ["Team A", "Team B"])     # labels of the groups
```

Example: processing times of two teams (exercise in the Boxplot section). `box_stats(shoe)` reproduces the shoe-size boxplot.

▶ **Try it yourself:** notebook, section 5 "Python Commands" – open in Colab

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Summary (1/4) – Empirical CDF

Definition

$$F(x) = \sum_{x_i \le x} f(x_i) \qquad \text{with values in } [0, 1]$$

Reading off quantiles

$$x_p = F^{-1}(p) = \min\{x \mid F(x) \ge p\}, \qquad 0 < p \le 1$$

Special case: $F^{-1}(0.5) = \tilde{x}$ (median); same as the position rule $x_{(\lceil n \cdot p \rceil)}$.

- Step function showing the cumulative relative frequency; jump height at x_i equals $f(x_i)$.
- Shares: $F(x)$ = share of values $\le x$; share in $(a, b] = F(b) - F(a)$.

Summary (2/4) – Histogram

Density of a class

$$\text{density}_j = \frac{f_j}{\Delta x_j} = \frac{\text{rel. frequency of class } j}{\text{class width}}$$

Area = relative frequency

$$\text{area}_j = \text{density}_j \cdot \Delta x_j = f_j, \qquad \sum_j f_j = 1$$

- Suitable for metric, classified data.
- Classes $[c_0, c_1]$, $(c_1, c_2]$, $\ldots$, $(c_{k-1}, c_k]$ with boundaries $c_0 < \ldots < c_k$.
- The choice of class width strongly affects the shape.
- Unequal class widths are possible – heights remain densities; the highest bar need not hold the most values.

Summary (3/4) – Boxplot

Five-number summary

$$x_{\text{min}} \text{ (or lower whisker)},\ x_{0.25},\ \tilde{x},\ x_{0.75},\ x_{\text{max}} \text{ (or upper whisker)}$$

$$\text{IQR} = x_{0.75} - x_{0.25}$$

Whisker bounds (Tukey, $k = 1.5$)

$$\text{Whisker}_{\text{min}} = \min\{x_i \mid x_i \geq x_{0.25} - k \cdot \text{IQR}\}$$
$$\text{Whisker}_{\text{max}} = \max\{x_i \mid x_i \leq x_{0.75} + k \cdot \text{IQR}\}$$

Outliers

All values $< x_{0.25} - 1.5 \cdot \text{IQR}$ or $> x_{0.75} + 1.5 \cdot \text{IQR}$ are displayed individually as outliers.

Summary (4/4) – Which Graphic for Which Question?

Question	Graphic	Read off
What share of the values is at most x? Up to which value lie at least $p \cdot 100\,\%$ of them?	Empirical CDF	$F(x)$, $F(b) - F(a)$, $x_p = \min\{x \mid F(x) \geq p\}$
What does the distribution look like (peaks, gaps, tails)?	Histogram	Height = density $f_j/\Delta x_j$, area = share f_j
Centre, spread, skewness, outliers – and comparison of groups?	Boxplot	Q_1, $\tilde{x}$, Q_3, IQR, whisker ends, points beyond the fences

Key takeaway

Course conventions: quantiles $x_{(\lceil n \cdot p \rceil)}$ and classes $(a, b]$, first class closed – in Python `method="inverted_cdf"` and `pd.cut(..., include_lowest=True)`.

Outlook: Part 3 – Statistical Inference

What we have so far

Parts 1 and 2 **describe** the data at hand – in numbers and in pictures.

Next part

We treat the data as a **sample** and infer back to the **population**: confidence intervals and hypothesis tests. The mean $\bar{x}$ and the standard deviation s of Part 1 become the building blocks of every interval and every test.

Key takeaway

From describing a sample to drawing conclusions about the population – honestly reporting the uncertainty.